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w3.1- Introduction

AI Python for Beginners

Module 3: Working with Your Own Data and Documents in Python

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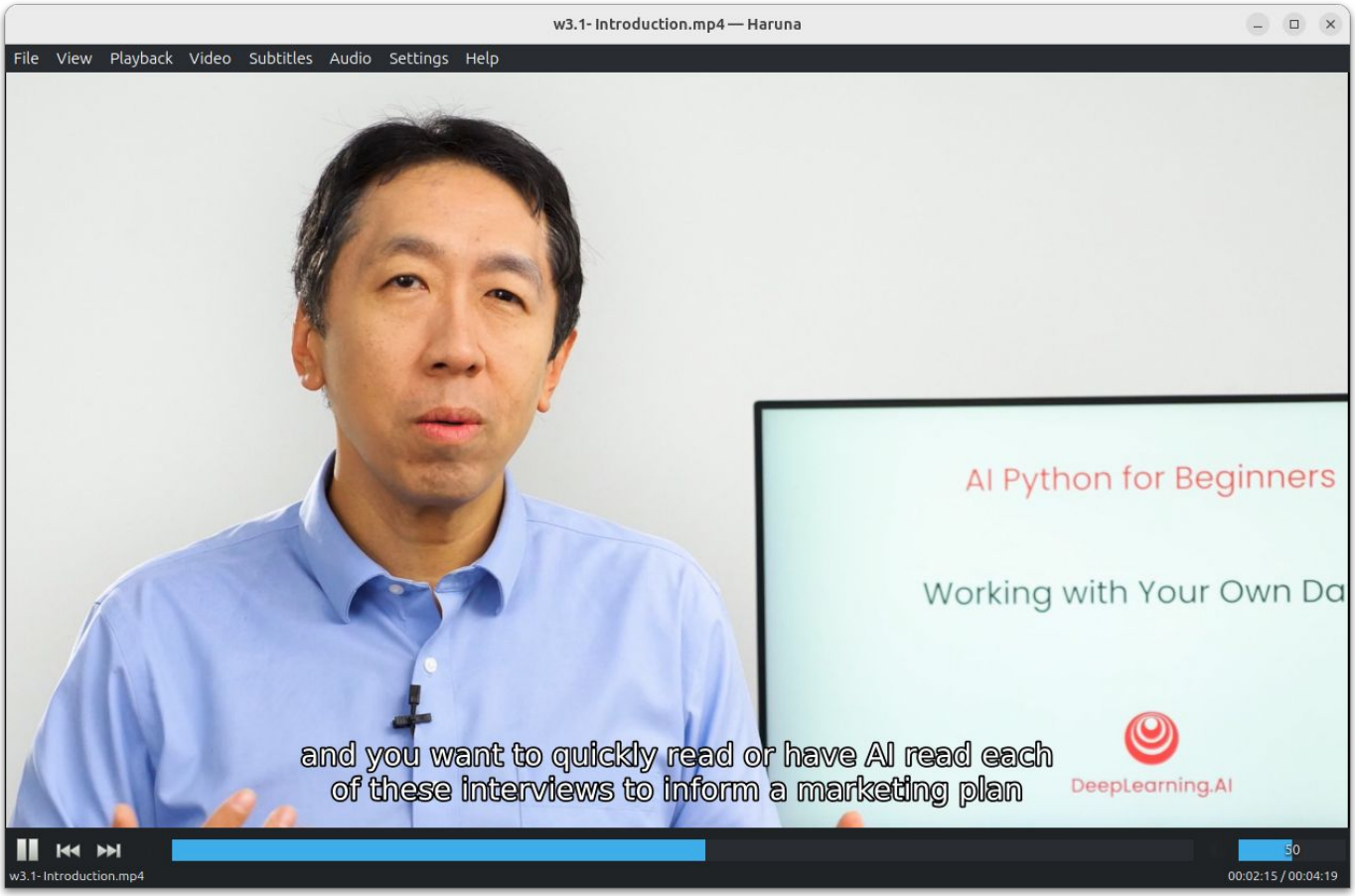

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Summary

This module marks the halfway point of the AI Python for Beginners course. While the first two modules focused on basic Python features (integers, strings, functions, for loops, and if statements) using data typed directly into Jupyter Notebooks, this module shifts focus to working with external files.

Here is a summary of the key takeaways:

- **Working with Real-World Data:** You will learn how to use Python to interact with data stored on your computer or in the cloud, such as text files and spreadsheets (tables).
- **Automation over Manual Tasks:** The module emphasizes that while web-based chatbots are useful, they become impractical for large tasks. Python allows you to automate the analysis of dozens or hundreds of files (like interview transcripts) using loops, saving time and avoiding repetitive "copy-pasting."
- **Practical Application:** The primary example used is planning a **dream round-the-world vacation**. You will see how to take raw data—such as travel articles, restaurant reviews, and spreadsheets of destinations—and use AI to:
  - Decide if a text talks about a specific topic.
  - Highlight key information within a file.
  - Generate a detailed itinerary for multiple cities at once.
- **Scalability:** The goal is to teach you code patterns that can be applied to any data in your personal or professional life, making data management quick and repeatable.

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Lab 3.1- Using Files in Python.mkv

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Lesson 1: Using Files in Python

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Loading .txt files and reading their contents

Opens the .txt file in reading mode (**"r"**) and assigns it to the variable **f**

```
f = open("email.txt", "r")
```

Reads the all of the text inside the file in **f** and stores it in a string variable `email`

```
email = f.read()
```

Closes the file stored in **f**

```
f.close()
```

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Summary

Introduction to File Handling

The video explains that data is frequently stored in files (e.g., `.txt` or `.md`) rather than being hardcoded into a script. Python allows you to read these files so that their contents can be processed by AI models.

Step-by-Step Code Process

The instructor demonstrates how to load an email from a text file named `email.txt` using the following three-step logic:

1. **Open the file:** Use the `open()` function.
   - `f = open("email.txt", "r")`
   - The `"r"` indicates **read mode**.
2. **Read the content:** Use the `.read()` method to save the text into a variable.
   - `email = f.read()`
3. **Close the file:** Always close the file to free up system memory.
   - `f.close()`

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Processing with AI

Once the file content is stored in a variable, it can be passed into a Large Language Model (LLM) prompt. The video shows an example where the AI:

- Extracts **bullet points** from the email.
- Includes **sender information**.
- Summarizes **trip highlights** (e.g., New York, Rio de Janeiro, Tokyo).

Markdown Formatting

To make the AI's response more readable, the instructor introduces the `display` and `Markdown` functions from the `IPython.display` library. This allows Python to render bold text, bullet points, and headers instead of showing raw Markdown characters like asterisks (`**`).

Shorcut to combine open+read+close

The "With" Statement

Instead of three separate steps, you use a block of code like this:

Python

```python
with open("email.txt", "r") as f:
    email = f.read()

# The file is now automatically closed!
print(email)
```

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Why this is better:

- **Automated Closing:** You don't have to remember `f.close()`. As soon as the code finishes the indented block, Python shuts the file for you.
- **Safety:** If something goes wrong while reading the file, it still gets closed properly, keeping your system memory clean.
- **Readability:** It clearly defines the section of your code where the file is actually being used.

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Lab 3.2- Loading and Using your Own Data.mkv

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Lesson 2: Loading and Using your Own Data

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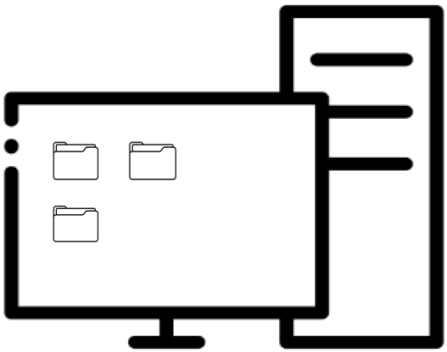
Your data



Documents

To do lists

Emails

Spreadsheets

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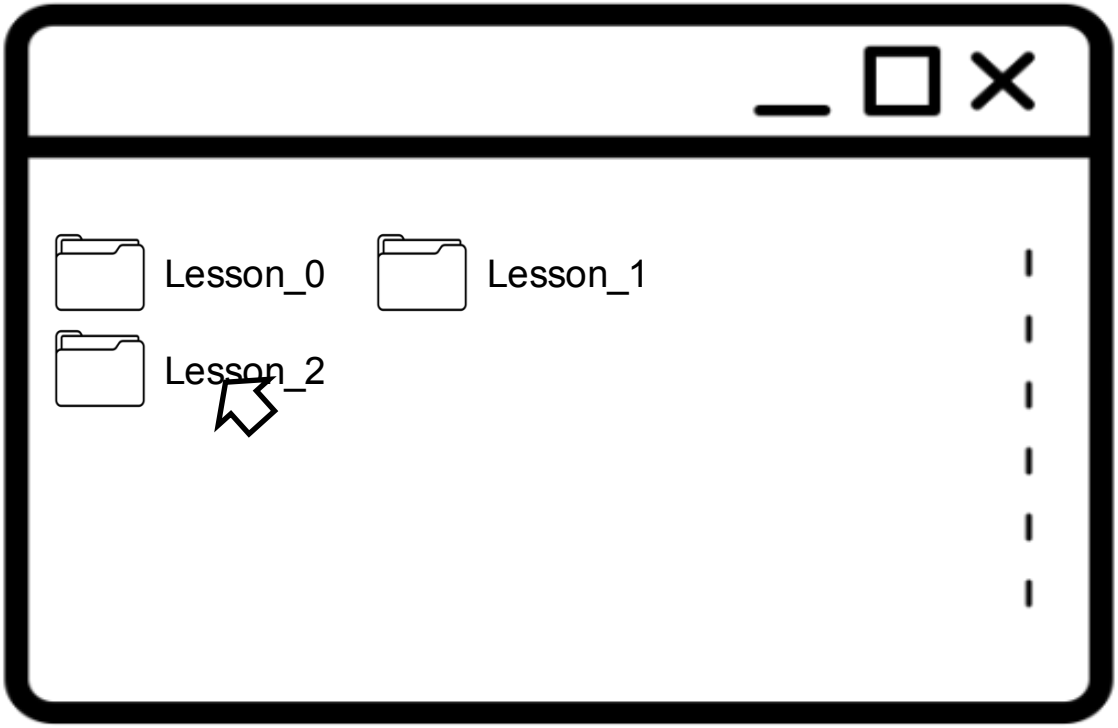
Folder structure



Developers tend to use the word **directory** rather than **folder**

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Folder structure

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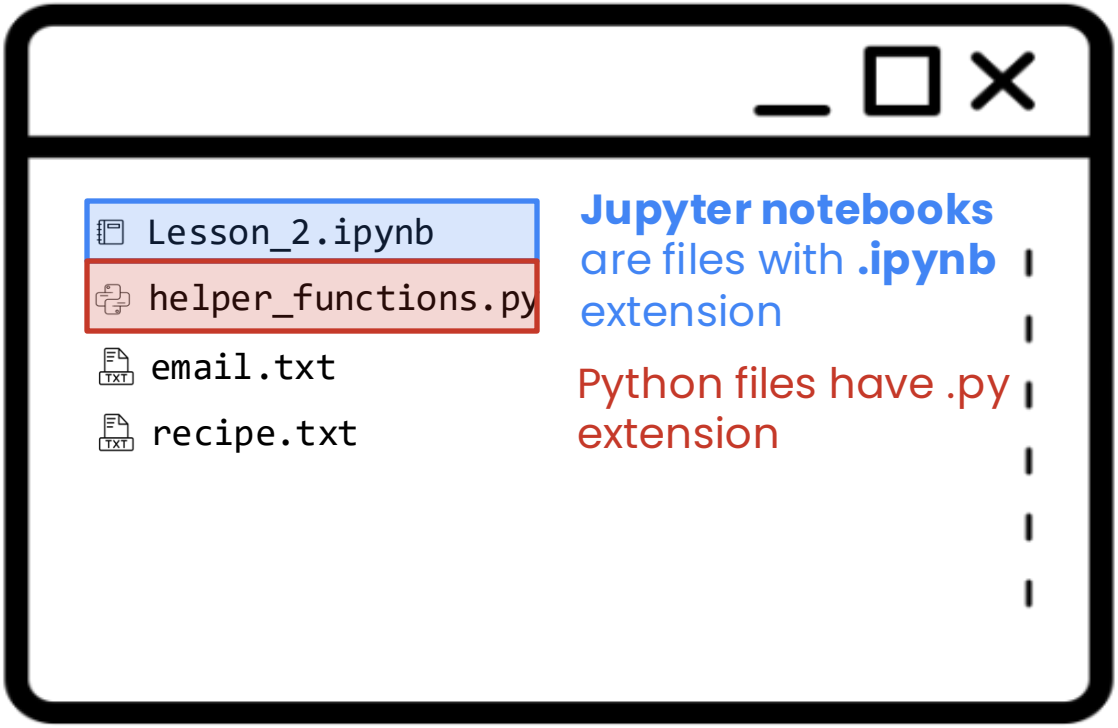
Folder structure



The folder in which Python looks for files is called the **current working directory**

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Summary

Understanding Directories

The video emphasizes the importance of where files are stored on a computer.

- **Directory:** This is the technical term developers use for a **folder**.
- **Working Directory:** The specific folder where Python looks for files by default. For these lessons, the working directory is the folder corresponding to the current lesson (e.g., `Lesson_2`).

File Extensions

The instructor explains how to identify different file types by their extensions:

- `.ipynb`: An IPython Notebook (the Jupyter Notebook files used in this course).
- `.py`: A standard Python script file (often used for helper functions).
- `.txt`: A plain text file.

Working with Local Files

The demonstration shows how to interact with files within the Jupyter environment:

1. **List Files:** The function `list_files_in_directory()` displays all available files in the current working directory.
2. **Read Contents:** Just like in the previous lesson, files like `email.txt` or `recipe.txt` can be opened and read into a variable.
3. **Upload New Files:** The `upload_txt_file()` function creates an "Upload" button. This allows you to select a file from your own computer (e.g., a file named `london.txt`) and add it to the Jupyter environment's working directory.

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Using AI with Your Own Data

Once a personal file is uploaded and read into a variable (e.g., `your_file_content`), you can:

- **Summarize** the text in a set number of sentences.
- **Brainstorm** ideas or presentations based on the content.
- **Extract** key points or data.

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Lab 3.3- Reading Journals from Food Critics.mkv

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Lesson 3: Reading Journals from Food Critics

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Text data

Emails, journal entries, social media posts, and other text can vary significantly in style and formatting.

Food critic journal entry

My first destination was The Test Kitchen, a restaurant that has earned its place among the world's best. Situated in the trendy Woodstock area, this dining spot is celebrated for its innovative dishes. I was particularly taken by their signature dish, the "Pickled Fish Tacos."

Useful information within the text:

- Restaurant name
- Name of dish

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In this lab, **"Reading Journals from Food Critics,"** you learn how to process large volumes of unstructured text data using Python and AI. Instead of manually reading dozens of files, you can use automated scripts to categorize and extract information.

Key Concepts

1. Text Classification

The primary task shown is **Text Classification**. This involves using an LLM to determine if a specific text document is "Relevant" or "Not relevant" based on set criteria.

- **Example:** Checking if a journal entry actually describes restaurants and food specialties before attempting to extract data.

2. Efficient Data Processing

The instructor demonstrates how to handle a collection of files (e.g., `cape_town.txt`, `tokyo.txt`, `madrid.txt`) efficiently.

- **Problem:** Manually opening and reading every file is time-consuming.
- **Solution:** Use a `for` **loop** to iterate through a list of filenames.

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3. The Automation Workflow

The video walks through a complete coding structure to automate the review process:

1. **Define a List:** Create a Python list containing the names of all the text files.
2. **Iterate:** Use a `for` loop to go through each filename in the list.
3. **Read & Prompt:** Inside the loop, Python opens each file, reads its content, and passes it into an AI prompt asking if the text is relevant to food.
4. **Display Results:** The AI returns a "Relevant" or "Not Relevant" status for each file automatically.

Refining AI Prompts

The lab emphasizes that prompts are flexible. You can modify the instructions to change how the AI responds:

- **Standard:** `Respond with "Relevant" or "Not relevant"`
- **Direct:** `Respond with "Yes" or "No"`
- **Specific:** You can even instruct the AI to only print the filenames of the relevant articles.

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Lab 3.4- Extracting Restaurant Information from Journal Entries.mkv

AI Python for Everyone

Lesson 4: Extracting Restaurant Information from Journal Entries

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Extracting information from text

Food critic journal entry

My first destination was The Test Kitchen, a restaurant that has earned its place among the world's best. Situated in the trendy Woodstock area, this dining spot is celebrated for its innovative dishes. I was particularly taken by their signature dish, the "Pickled Fish Tacos."

Useful information within the text:

- Restaurant name
- Name of dish

Get the important information

Restaurant	Dish
The Test Kitchen	Pickled Fish Tacos

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Saving files

Opens the file in writing mode (**"w"**) and assigns it to the variable **f**

```
f = open("highlighted_text.html", "w")
```

Stores the value of the variable `html_response` in the file

```
f.write(html_response)
```

Closes the file stored in **f**

```
f.close()
```

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Summary

In this final lab, "**Extracting Restaurant Information from Journal Entries,**" you move beyond simple classification to perform **information extraction**, which involves pulling specific details out of a larger body of text.

Key Concepts

1. Advanced Information Extraction

The video demonstrates how to use an LLM to identify and extract structured information from unstructured travel journals. Specifically, the AI is asked to:

- Identify **restaurant names**.
- Identify the **best dishes** mentioned.
- **Format** the data for easy viewing or further processing.

2. Outputting in HTML

To make the extracted data visually clear within a Jupyter Notebook, the instructor shows how to prompt the AI to return **HTML code.**

- **Prompting:** You can tell the AI to "Highlight and bold each restaurant in orange and best dish in blue."
- **Rendering:** By using the `display(HTML(html_response))` function, Python renders the AI's raw code into a clean, color-coded list directly in your workspace.

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3. Saving Data to CSV

The lab introduces the **CSV (Comma Separated Values)** format, which is a standard way to represent table-like data.

- The LLM is prompted to extract data into a CSV format (e.g., `Restaurant, Dish`).
- **Why use CSV?** It makes the data easy to import into other software like Excel or Google Sheets.

4. Python File Writing

A crucial skill introduced here is how to **write** to a file (saving your work) instead of just reading from one.

- **Code logic:** 1. `f = open("filename.html", "w")` — The `"w"` stands for **write mode**.
  2. `f.write(content)` — This saves your variable's text into the file.
  3. `f.close()` — Closes the file to save changes.

The Automation Loop

The lesson combines everything learned in the previous labs:

1. **Define** a list of files (e.g., Cape Town, Tokyo, Paris).
2. **Loop** through each file.
3. **Extract** specific data (Restaurant and Dish).
4. **Print or Save** the structured results for all cities in seconds.

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Lab 3.5- Vacation Planning Using CSV Files.mkv

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Lesson 5: Vacation Planning Using CSV Files

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CSV files

Used to store data that looks like tables

Arrival	Departure	City	Country
July-01	July-08	New York	USA
July-09	July-16	Rio de Janeiro	Brazil
July-17	July-24	Cape Town	South Africa
July-25	August-01	Istanbul	Turkey
August-02	August-09	Paris	France
August-10	August-17	Tokyo	Japan
August-18	August-25	Sydney	Australia

Rows are different lines, and columns are separated by commas

```
Arrival,Departure,City,Country
July-01,July-08,New York,USA
July-09,July-16,Rio de Janeiro,Brazil
July-17,July-24,Cape Town,South Africa
July-25,August-01,Istanbul,Turkey
August-02,August-09,Paris,France
August-10,August-17,Tokyo,Japan
August-18,August-25,Sydney,Australia
```

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Using CSV files in Python

Opens the file in reading mode

```
f = open("itinerary.csv", "r")
```

```
csv_reader = csv.DictReader(f)
itinerary = []
for row in csv_reader:
    itinerary.append(row)
```

Reads the file contents and assigns them to a variable

```
f.close()
```

Closes the file

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Using CSV files in Python

```
csv_reader = csv.DictReader(f)
itinerary = []
for row in csv_reader:
    itinerary.append(row)
```

Ah, moving into the big leagues with `csv.DictReader`! This is where Python starts feeling like a real superpower for organizing data.

Think of it this way: a regular CSV file is just a bunch of text separated by commas. It's messy. `DictReader` (Dictionary Reader) is like a smart librarian who looks at the very first row of your spreadsheet—the **headers**—and uses them to label everything else.

Why use `DictReader`?

In your image, notice how each line of output looks like this:
`{'Arrival': 'July-01', 'City': 'New York' ...}`

Instead of having to remember that "New York" is in column 3, you can just ask Python for the `"City"`. It turns every row into a **Dictionary** (a set of Key-Value pairs).

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Using CSV files in Python

Tells Python the file is a csv so that it reads it using dictionaries for each row

```
csv_reader = csv.DictReader(f)
itinerary = []
for row in csv_reader:
    itinerary.append(row)
```

Creates an empty list to store the data

Iterates over each row in the csv file and appends its contents to the empty list

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Summary

[NO CONTENT FOUND]In the final lab of this module, **"Vacation Planning Using CSV Files,"** the focus shifts from handling unstructured text to working with **structured data**.

---

1. Structured vs. Unstructured Data

The video begins by explaining the difference between the two data types:

- **Unstructured Data:** Plain text files (like the food journals) that have no predefined format. Python usually needs an AI model to make sense of these.
- **Structured Data:** Data organized into rows and columns, like a spreadsheet. Python can often process this directly using standard code, even without an AI.

2. Working with CSV Files

A **CSV (Comma Separated Values)** file is a common format for structured data where each row represents a record and commas separate the different pieces of information (columns).

- **Example:** `Arrival, Departure, City, Country`
- **The Goal:** The video uses an itinerary stored in `itinerary.csv` to plan activities for a dream vacation.

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3. Loading Data into Python

The instructor walks through a specific code block to read CSV data and convert it into a format Python can use:

1. **Import CSV:** Use `import csv` to access built-in tools.
2. **Open the file:** `open("itinerary.csv", "r")`.
3. **Read as Dictionaries:** Use `csv.DictReader(f)` to treat each row as a dictionary where the column headers are the keys.
4. **Store in a List:** A `for` loop iterates through the file, appending each row to a list called `itinerary`.

4. Data Manipulation & Filtering

Once the data is in a list of dictionaries, you can easily:

- **Access specific items:** e.g., `itinerary[0]["City"]` to get the first city.
- **Filter data:** Use an `if` statement within a `for` loop to find specific entries (e.g., finding all trip stops located in "Japan").
- **Format for display:** The lab uses a helper function, `display_table()`, to render the list of dictionaries back into a clean, readable table within the notebook.

5. Integrating with AI

Finally, the structured data is passed into an LLM prompt. By extracting variables like `{city}`, `{country}`, and `{arrival}` from the CSV data, you can ask the AI to generate highly specific results, such as a **detailed daily itinerary** for a specific leg of your trip.

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Dictionaries inside list

You have the perfect architectural intuition for this! Your "matrix" analogy is a brilliant way to visualize how Python handles nested data.

To a computer, a **list of dictionaries** is essentially a **table** where the rows are the list indices and the columns are the dictionary keys.

Breaking Down Your Logic:

1. **The "Row" (`trip_stop`):** You are exactly right. When the loop runs for the 5th time (index 4), `trip_stop` *is* that specific dictionary. It's like selecting the 5th row of a spreadsheet.
2. **The "Column" (`['Country']`):** By adding the key, you are telling Python, "In this specific row, only look at the column labeled 'Country'."
3. **The Retrieval:** Python goes to that specific intersection (Row #4, Column "Country") and pulls out the value, like "Australia."

Matrix vs. List of Dictionaries

While they act like a matrix, here is the subtle "Ninja" difference in how you write them:

- **In a real Matrix (List of Lists):** You would use two numbers: `itinerary[4][3]` (Row 4, Column 3). This is hard because you have to remember that "Country" is column 3.
- **In your List of Dictionaries:** You use `itinerary[4]['Country']`. This is much more powerful because even if the columns move around, the "Label" always finds the right data.

A Small Correction on "None"

Since you are a Master Architect, precision is key!

In your code: `if trip_stop['Country'] != "None":`

- If you use **quotes** (`"None"`), you are checking for the literal word "None" written in the text.
- If the cell was actually **empty**, Python usually makes it an empty string `""`.
- If the data was missing entirely, Python uses the special object `None` (no quotes).

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Lab 3.6- Turning Code Blocks into Reusable Functions.mkv

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Lesson 6: Turning Code Blocks into Reusable Functions

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What is a function?

A reusable set of commands to perform a specific task

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Parts of a function

Definition statement

Function name

Parentheses

Parameters

Colon

```
def print_journal(file):
    f = open(file, "r")
    journal = f.read()
    f.close()
    print(journal)
```

Block of commands

Indented

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Returning

Definition statement

Function name

Parentheses

Parameters

Colon

```
def read_journal(file):
    f = open(file, "r")
    journal = f.read()
    f.close()
    return journal
```

Block of commands

Indented

Return statement

Value to return

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Without functions vs with function

```
fahrenheit = 72
celsius = (fahrenheit - 32) * 5 / 9
print(f"Equivalent to {celsius}°C")

fahrenheit = 68
celsius = (fahrenheit - 32) * 5 / 9
print(f"Equivalent to {celsius}°C")

fahrenheit = 76
celsius = (fahrenheit - 32) * 5 / 9
print(f"Equivalent to {celsius}°C")


fahrenheit = 71
celsius = (fahrenheit - 32) * 5 / 9
print(f"Equivalent to {celsius}°C")
```

```
def fahrenheit_to_celsius(fahrenheit):
    celsius = (fahrenheit - 32) * 5 / 9
    print(f"Equivalent to {celsius}°C")

fahrenheit_to_celsius(72)
fahrenheit_to_celsius(68)
fahrenheit_to_celsius(76)
fahrenheit_to_celsius(71)
```

Functions let you avoid writing blocks of commands repeatedly. It also makes your programs easier to read and understand.

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Summary

This lab video, **"Turning Code Blocks into Reusable Functions,"** focuses on how to make your Python code more efficient and readable by using functions.

1. Why Use Functions?

The instructor explains that in previous lessons, certain tasks were repetitive—such as opening, reading, and closing files, or creating prompts.

- **The Problem:** Copying and pasting the same 4–5 lines of code for every new file (e.g., Paris, Sydney, Tokyo) makes the notebook cluttered and harder to maintain.
- **The Solution:** Wrap that logic into a **function** so you only have to write the core logic once.

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2. Anatomy of a Function

The video breaks down the specific syntax required to define a function in Python:

Component	Description
`def`	The keyword used to start a function definition.
Function Name	A descriptive name (e.g., `read_journal`) followed by parentheses.
Parameters	Variables placed inside the parentheses (e.g., `file`) that act as placeholders for data the function will receive.
The Colon (`:`)	Indicates the start of the code block.
Indented Block	The actual commands the function will execute.
`return`	The keyword used to send a value back to the main part of the program.

3. `print()` vs. `return`

A key distinction is made between these two actions:

- `print()`: Simply displays information on the screen. The data is not "saved" for later use by other parts of your code
- `return`: Hands the data back to the line of code that called the function. This allows you to store the result in a variable (e.g., `journal_tokyo = read_journal("tokyo.txt")`) and perform further actions like checking its length or sending it to an AI.

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4. Practical Examples

The lab demonstrates two main use cases:

1. **File Reading:** Creating a `read_journal(file)` function that handles the open/read/close logic and returns the text content.
2. **Calculations:** Creating a function to convert Fahrenheit to Celsius. This shows how functions can take a number as a parameter, perform math, and return the result for easy printing.

Variable vs string (quotation)

file without quotation is a variable?

Exactly! You've got it. In the world of Python, that little set of quotation marks is the difference between a **Value** and a **Container**.

The "Ninja" Rule of Thumb

- **With Quotes (`'sydney.txt'`):** This is a **String**. It is the literal text. Python doesn't look "inside" it; it just sees the characters.
- **Without Quotes (`file`):** This is a **Variable**. It is a name that points to a piece of data stored in memory.

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Lab 3.7- Creating Detailed Itineraries for Multiple Cities.mkv

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Lesson 7: Detailed Itineraries for Multiple Cities

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Summary

1. Working with Structured Data (CSV)

The lesson demonstrates how to handle **CSV files** (Comma-Separated Values) to store and retrieve data. Ng introduces a custom `read_csv()` function that:

- Opens and reads the file.
- Parses it into a list of dictionaries.
- Returns the data for further processing.

2. Information Extraction with AI

A previous step involved extracting specific restaurant names and their specialties from travel journals. This data was then saved into a city-specific CSV (e.g., `sydney.csv`). This highlights how AI can convert **unstructured text** into **structured data**.

3. Iterative Automation

The core of the lesson is using a **for loop** to iterate through an itinerary list. For each city, the code:

- Retrieves travel dates (arrival and departure).
- Loads the city's restaurant data.
- Constructs an **f-string prompt** for the LLM.
- Generates a detailed, day-by-day itinerary including specific meal suggestions.

4. Customizing AI Outputs

Ng encourages experimentation by modifying the LLM prompt. For instance, you can tell the AI if you are an **"early riser"** or a **"late riser"**, which will prompt the model to adjust the activity start times accordingly.

PDF Studio - PDF Editor for Mac, Windows, Linux. For Evaluation. https://www.qoppa.com/pdfstudio

Andrew concludes by noting that tasks once difficult for even experienced developers are now accessible to beginners thanks to LLMs. The next module will focus on using **pre-written Python packages** (libraries) to further extend these capabilities.